

2. Endogenetic Movements

Std 9th

Sub – Geography

Q 1. Tick in front of the correct option ✓

(a) On which of the following are slow movements in the earth's interior dependent?

Ans. Velocity

(b) When waves divert from each other, what do they create

Ans. Tension

(c) For the formation of a rift valley, which of the following processes should occur in the earth's crust?

Ans. Tension

(d) Which of these is a fold mountain?

Ans. The Himalayas

(e) The formation of extensive plateaus is a result of which type of movements ?

Ans. Continent-building

Q 2. Give geographical reasons.

(a) Buildings collapsed at the foothills of the Himalayas because of an earthquake. Before collapsing they were moving forward and backward.

Ans. 1. The seismic waves can be divided into primary, secondary and surface waves.

2. The primary waves are the first ways to reach the surface of the earth as energy is released inside the earth's interior.

3. Subjected to 'P' wave the particles of the rock move in the direction of the wave to and fro therefore they are also known as forward and backward waves.

4. Thus, the buildings at the foothills of the Himalayas move forward and backward as they subjected to P wave.

(b) There is a difference in the formation of the Meghalaya Plateau and the Deccan Plateau.

Ans. 1. The Meghalaya plateau was formed due to slow movement (orogenic movements) where is the Deccan Plateau was formed due to sudden movement (fissure volcano).

2. During the formation of Meghalaya plateau a part of the earth's crust between two parallel faults was lifted forming the Meghalaya plateau where is the Deccan Plateau was formed due to molten material (Lava) coming out with the eruption on both the sides of the fissure volcano.

3. Thus there are difference in the formation of Meghalaya and Deccan Plateau.

(c) Most of the volcanoes are found on the plate boundaries.

Ans. 1. The plate boundaries are directly related to the area of volcanoes.

2. Most of the volcanoes are located in the plate of boundaries.

3. Part of the boundaries with slide into the earth's crust results in loss of minerals and such boundaries are called as plate consuming boundaries in the areas when new material (magma) is coming onto the earth's crust they are called as plate creating boundaries.

4. Both of the processes results in the earthquake and volcanic eruptions.

Thus most of the volcanoes are situated at the plate boundaries.

(d) The Barren Island is becoming conical in shape.

Ans. 1. Barren Island which was considered a dormant volcano has become active since February 2017.

2. Mainly dust, smoke and mud are coming out of this eruption along with some lava.

3. During eruptions, the lava is spread around the mouth of this vent, when it comes out.

4. As a result, a cone shaped mountain starts forming therefore, barren island is becoming conical shape.

(e) Volcanic eruptions can cause earthquakes.

Ans. 1. Generally, the movements in the earth's interior occur in the upper layer of the mantle.

2. Tremendous energy is released by radioactive minerals present here.

3. This leads to sudden movement in the form of volcanic eruption where hot solid, liquid and gaseous materials are forcefully thrown out from the mantle of the Earth on to the surface of the earth.

4. This causes a lot of instability the within the crust of the Earth which would also lead to the release of pressure in the form of seismic waves or tremors causing earthquake.

Q 3. Identify and name the internal movement.

(a) Tsunamis are generated in coastal areas.

Ans. Sudden

(b) The Himalayas are an example of fold mountains.

Ans. Slow

(c) Molten magma is thrown out of the earth's mantle.

Ans. Sudden

(d) Rift valley is formed because of faulting.

Ans. Slow

Q 4. Arrange the following statements in chronological order in which an earthquake occurs.

Ans. (b) The plates suddenly move.

(c) Due to the movements in the mantle, compression goes on increasing.

(d) Along the weak points (faultlines) rocks break apart.

(e) Stored energy is released in the form of seismic waves.

(a) The earth's surface vibrates

Q 5. Distinguish between

(a) Block Mountain and Fold Mountain

Ans.

Block Mountain	Fold Mountain
1. When a part of the earth crust between two parallel faults is lifted up leads to forming a mountain.	1. Because of the energy waves and pressure working towards each other soft layers of rock leads to fold.
2. It is called as block mountain.	2. It leads to the formation of fold mountain.
3. Block mountains are formed due to tension.	3. Fold mountains are formed due to compression.
4. Eg. The Black Forest mountain of Europe and Meghalaya plateau in India.	4. Eg. The Himalayas, the Aravalis, the Alps, the Andes and the roboties.

(b) Primary and Secondary Seismic Waves

Ans.

Primary waves	Secondary Waves
1. Primary waves are the first waves to reach the surface of the earth when energy is emitted in the interior of the earth.	1. The waves which reach the surface of the earth after primary waves are called secondary waves.
2. Primary waves can travel through solid, liquid and gaseous state.	2. Secondary waves can travel only through solid medium.
3. Primary waves changes direction while travelling through liquid medium.	3. It gets absorbed as it enters the liquid state
4. The building on the surface of the earth move to and fro because of the primary wave.	4. The building on the surface of the earth move up and down because of the secondary waves.

(c) Earthquakes and volcanoes

Ans.

Earthquakes	volcanoes
1. Earthquake is the movement in the earth's crust.	1. A volcano is an opening in the earth's interior from which hot ,solid, liquid and gaseous materials are thrown out from the mantle of the earth.
2. Earthquake in ocean can cause tsunamis.	2. Volcanic eruption in the ocean can cause earthquake.
3. The seismic waves can be classified into three waves primary, secondary and surface waves.	3.The volcanoes can be classified into three types dormant volcanoes, active volcanoes and extinct volcanoes.
4. Nothing is ejected out during an earthquake.	4. During an volcanic eruption ash, water vapour, various types of poisonous and inflammable gas is and molten magma is emitted from interior of the earth.

Q 6. Answer in brief

(a) Give reasons why an earthquake occurs.

Ans. When Energy is released in the interior of the earth, tremendous tension is created in the earth's crust. When the tension goes beyond limits, the energy is released in the form of waves. This results in trembling of the earth's surface, i.e. earthquake occurs.

Causes of earthquakes :

1. Moving of the plates
2. Colliding of plates
3. Plates sliding one below the other

4. Forming of fractures in rock layers due to tension in the interior of the earth.

5. Occurring of volcanic eruptions.

(b) Which type of movements have led to the formation of the major fold mountains in the world?

Ans. 1. Fold mountains are formed due to mountain building ~~movement~~. (orogenic movement)

2. Because of these energy waves and pressure working towards each other and in horizontal direction, the layers of the soft rocks form folds.

3. If the pressure is very high, largescale folds are formed and their complexity increases.

4. As a result, the surface of the earth gets uplifted and fold mountains are formed.

5. The Himalayas, the Aravalis, the Rockies, the Andes, the Alps are the major fold mountains of the world.

(c) How is the magnitude of the earthquake related to the collapse of houses

Ans. 1. Through a Seismo graph showing movement of seismic waves (Seismograph) can be generated.

2. After studying this graph, the magnitude of the earthquake is known.

3. The magnitude of the earthquake is measured with the help of Richter scale.

4. It is because of the primary waves the building on the earth's surface moved forward and backward.

5. Because of the secondary waves buildings on the earth's surface moved up and down.

(d) What are the effects of earthquakes on the earth's surface and human life?

Ans. Effects of earthquake :

1. Cracks /fractures develop on the ground.

2. Causes landslides which leads to sliding of rocks.

3. Sometimes the groundwater changes its course. For example, wells may get water or may dry up.

4. Some areas get uplifted while some may subside.
5. Tsunamis are generated in oceans. These waves can cause great loss of life and property in the coastal areas.
6. In snow-covered areas, avalanches may occur.
7. Buildings collapse and loss of life and property occurs.
8. Transportation routes get disrupted.
9. Communication system collapses.

(e) Explain the types of seismic waves.

Ans. The seismic waves can be divided into primary, secondary and surface waves.

1. Primary or 'P' waves: i. These are the first ones to reach the surface of the earth after the energy is emitted in the earth's interior.

ii. They travel at a very fast speed from the focus of the earthquake in radial direction.

iii. This wave moves to and fro, therefore these waves are also called forward-backward waves. These waves can travel through all the three states - liquid, solid and gaseous.

iv. It is because of the primary waves that the buildings on the earth's surface move back and forth.

2. Secondary or 'S' waves : i. The waves which reach the earth's surface after the primary waves are called secondary waves.

ii. These waves also scatter in all directions from the focus of the earthquake.

iii. Their velocity is lesser than the P waves. The particles lying in the way of these waves move up and down in the direction of energy transfer.

iv. These waves can travel only through the solid medium. They get absorbed as they enter the liquid medium.

v. The buildings on the earth's surface move up and down because of these waves.

vi. These are more destructive than the P waves.

(3) Surface or 'L' waves : i. These waves are generated after the main P and S waves reach the epicenter.

ii. They travel in the direction of the circumference of the earth along the crust. They are highly destructive.

(f) Explain the types of volcanoes on the basis of periodicity of eruption with examples.

Ans. According to the periodicity of the volcanic eruptions, three types of volcanoes can be identified:

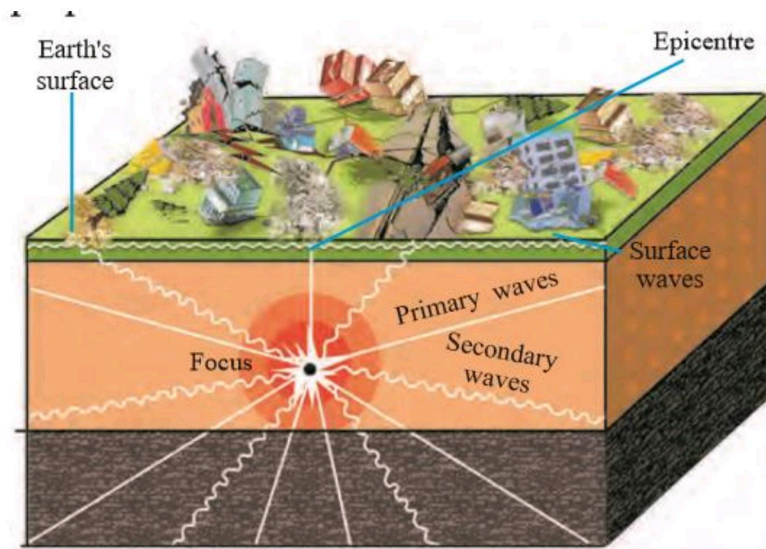
1. Active volcano If the volcanic eruptions are regular even in the present times, then such volcanoes are called active volcanoes. For example, Mt Fujiyama in Japan, Mt. Stromboli in Mediterranean Sea.

2. Dormant volcano When a volcano has not erupted since long, but may become active suddenly it is called dormant volcano. For example, Mt. Vesuvius in Italy, Mt. Katmai in Alaska, Barren Island, India.

3. Extinct volcano Those volcanoes which have not erupted in the past since long and are not likely to erupt in the future are called extinct volcanoes. For example Mt. Kilimanjaro in Tanzania.

Q 7. Show the epicentre, focus and the primary, secondary and surface waves of an earthquake with the help of a neat labelled diagram.

Ans.



Q 8. Show the following on a given outline map of the world.

(a) Mt. Kilimanjaro

(b) Mid-Atlantic Earthquake zone

(c) Mt. Fuji

(d) Krakatoa

(e) Mt. Vesuvius